

Reese Ludwick

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EDUCATION

Bachelor's of Science (BS)

Computer Science (GPA in Major - 3.43)

Shepherd University

August 2021 - May 2025

EXPERIENCE

Software Associate

Remote

Cognizant Technology Solutions

December 2025 - Present

- Develop and maintain **GenAI and agentic AI systems** using LangChain, LangGraph, and Python, contributing to R&D initiatives and client-facing deliverables.
- Design and implement **RAG pipelines and MCP server integrations** to connect enterprise tools and language models for production-ready workflows.
- Collaborate with cross-functional teams to scope, prototype, and deliver **AI-powered data analysis tools** that drive measurable business insights.

Freelance AI Engineer & Full-Stack Developer

Remote

Independent Contract (WorldCovers / APMC)

May 2026 - Present

- Engaged as a paid contract engineer to maintain and enhance multiple production web properties (chinaoverprints.com, ianthom.org) and contribute to **WorldCovers**, a philatelic cataloging platform sponsored by the U.S. Philatelic Classics Society.
- Deliver features and automation against a **14-table relational PostgreSQL schema** powering a **Django REST API backend** with a **React / TypeScript frontend**.
- Leverage **AI-augmented engineering (Claude Code)** to scaffold, build, and deploy production-quality web work on compressed timelines, coordinating directly with project stakeholders and collaborators.

GenAI Developer Contractor

Remote

Cognizant Technology Solutions

June 2025 - December 2025

- Built and delivered **GenAI proof-of-concept solutions** using LangChain, LangGraph, and Python, accelerating team adoption of agentic AI patterns.
- Applied **prompt engineering and model evaluation techniques** to optimize LLM outputs across various business use cases.
- Built **custom FastMCP servers** and deployed **external MCP integrations** alongside **AWS infrastructure** to equip agents with dynamic tool access, enabling modular, production-style agentic workflows.

IT Student Helper

Hedgesville, WV

James Rumsey Technical Institute

August 2019 - May 2021

- Resolved over **300+ support tickets** involving network connectivity, printing issues, and hardware repairs.
- Contributed to Wi-Fi overhaul that improved connection stability and cut latency by **25%** across campus.
- Configured and tested **over 100 AV systems** and networking equipment to support seamless academic instruction.
- Documented network layout and IP assignments, increasing efficiency in IT support responses for **100% of operations**.

PROJECTS

CFO Agent — Personal Finance AI Platform | Python, FastAPI, LangGraph, PostgreSQL, Redis, Anthropic API, Plaid

Independent | 2025-2026

- Built a **personal-finance AI agent** on a FastAPI + LangGraph backend with PostgreSQL persistence and Redis caching, integrating **Plaid** for automated financial account aggregation.
- Designed a **hybrid context architecture** combining automated aggregation with a manual data “vault” to deliver genuinely personalized financial reasoning; architected for multi-tenant productization.

CreatorClip — AI Video Auto-Clipper | Python, FastAPI, Celery, pgvector, WhisperX, Voyage AI, Anthropic API, ffmpeg,

YouTube OAuth

Independent | 2025-2026

- Engineered an **AI video-clipping pipeline** that learns a creator's channel style via a custom “DNA” module, automating short-form clip generation from long-form YouTube content.

- Built an **asynchronous processing pipeline** (Celery workers) spanning transcription (WhisperX), embeddings (Voyage AI + pgvector), and automated rendering (ffmpeg), with full YouTube OAuth integration.

Agentic Software Development Team | Python, Langchain, LangGraph, Amazon Bedrock, ML Models

Cognizant R&D | 2025

- Architected a **multi-agent software development system** using LangGraph with specialized agents for code generation, testing, security review, and supervision - automating the full software build lifecycle end-to-end.
- Integrated **Amazon Bedrock and open-source ML models** as inference backends, enabling flexible model selection across agent roles within the pipeline.

Customer Call Data Analysis & Insights Engine | Python, NLP, Sentence Transformers, Embeddings, Prompt Engineering

Cognizant R&D | 2025-2026

- Applied **NLP clustering with sentence transformers and embeddings** to categorize large volumes of customer call data into semantically meaningful issue groups.
- Leveraged **Agentic AI to surface niche call consolidation patterns**, delivering actionable executive reporting on unique customer issue trends.

TECHNICAL SKILLS

Programming Languages - Python, SQL

DevOps & System Administration - Linux, Git, Github, Azure Fundamentals, Ubuntu

Artificial Intelligence and Machine Learning - LangChain, LangGraph, RAG Pipelines, MCP Servers, Amazon Bedrock, Sentence Transformers, TensorFlow, Pandas, Numpy, pgvector

Tools & Workflow - VS Code, Claude Code, Docker, FastAPI, PostgreSQL, Redis, Celery, Jupyter Notebook, React / TypeScript, Django REST

CERTIFICATIONS

TensorFlow Developer Specialization - [DeepLearning.ai](https://www.deeplearning.ai)

Linux Certification - Boot.dev

Git Certification - Boot.dev

Microsoft Certified: Azure Fundamentals (AZ-900)

Microsoft Office Specialist: Microsoft Word 2016

Github Copilot: Cognizant Technology Solutions Learning Center

Telecom: Cognizant Technology Solutions Learning Center

Generative AI Context Engineering: Cognizant Technology Solutions Learning Center